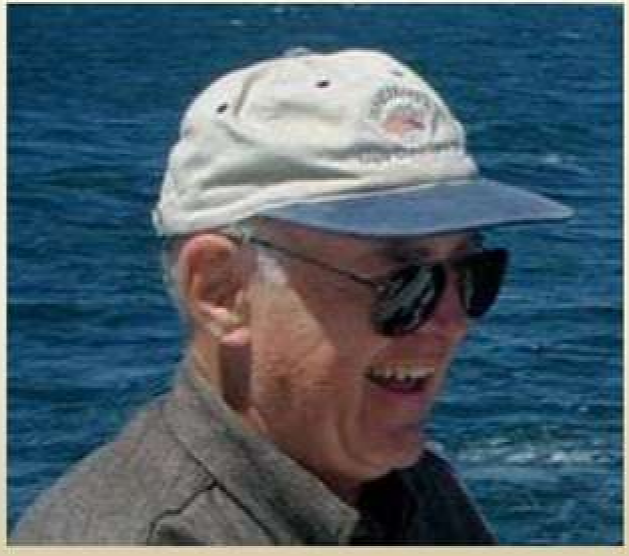
What is Moore's Law ?



Mr. Gordon Moore

Moore's law is the observation that,

over the history of computing hardware, the number

of transistors on integrated circuits doubles

approximately every two years.

The law is named after Intel co-founder Gordon E. Moore,

who described the trend in his 1965 paper.

His prediction has proven to be accurate, in part because the law is now used in the semiconductor industry to guide long-term planning and to set targets for research and development.

The capabilities of many digital electronic devices are strongly linked to Moore's law: processing speed, memory capacity, sensors and even the number and size of pixels in digital cameras.

All of these are improving at (roughly) exponential rates as well. This exponential improvement has dramatically enhanced the impact of digital electronics in nearly every segment of the world economy.

Moore's law describes a driving force of technological

and social change in the late 20th and early 21st

centuries.

The period often quoted as "18 months" is due

to Intel executive David House, who predicted that period

for a doubling in chip performance (being a combination of

the effect of more transistors and their being faster).

Although this trend has continued for more than half a century, Moore's law should be considered an observation or conjecture and not a physical or natural law.

Sources in 2005 expected it to continue until at least 2015 or 2020.However, the 2010 update to the International Technology Roadmap for Semiconductors has growth slowing at the end of 2013

…after which time transistor counts and densities

are to double only **every three years.**